

Limit Order Book Execution Plan

Algo Type1 Signals and Internal Flow Optimization

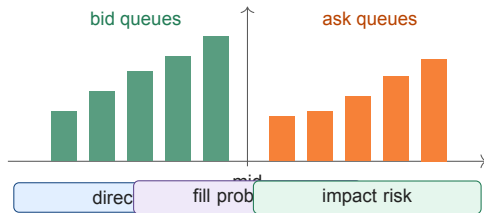
15-minute leadership briefing | June 4, 2026

■ One-Page Context

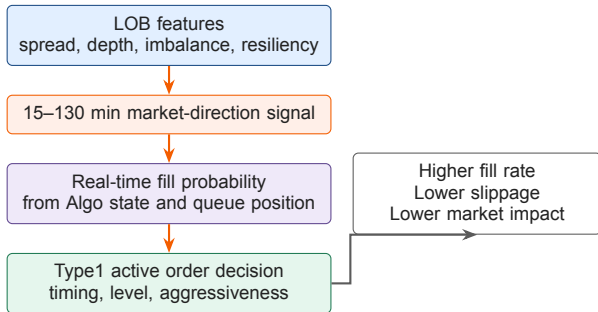
Why LOB matters

The order book is the closest observable state before execution: price direction, queue depth, fill probability, and impact risk are visible together.

- ▶ Keep the technical introduction to one page.
- ▶ Use LOB features to improve execution, not only to predict price.
- ▶ Focus on two deliverables: **Algo Type1** and **department-level internal flow**.



■ Output 1: Algo Type1 Execution Alpha



Business use

Support the current Algo instance for
Type1 active order placement.

- ▶ Signal horizon: 15, 30, 60, 90, 130 minutes.
- ▶ Combine market direction with execution probability.
- ▶ Improve fill probability when direction is favorable.
- ▶ Avoid chasing in adverse LOB regimes.

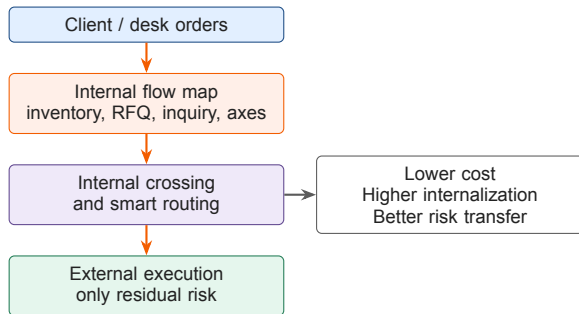
■ Output 1: Delivery Plan and KPIs

Prototype path

1. Build LOB feature stream and direction labels.
2. Train interpretable baseline signals by horizon.
3. Join signal with Algo fill-probability engine.
4. Run shadow trading, then controlled live A/B test.

KPI	Measurement
Fill rate uplift	Type1 fills vs. baseline by regime
Slippage saving	bps vs. arrival / decision price
Impact control	post-trade adverse move
Model stability	hit rate and IC across horizons
●	●
Data	Shadow
	A/B
	Scale

■ Output 2: Department Internal Flow Engine



Department-level target

Develop an internal flow optimization layer that reduces transaction cost through internal crossing, better routing, and pre-trade liquidity selection.

- ▶ Convert historical flow into actionable liquidity intelligence.
- ▶ Match compatible internal demand before external trading.
- ▶ Use LOB and flow signals to choose execution timing.

■ Value Case: RMB 90mm Target

Fixed-income approximation

$$\Delta \text{PnL} \approx \text{Notional} \times \text{Duration} \times \text{Saving}_{bp} \times 10^{-4}$$

- ▶ 2025 volume: **RMB 45tn**.
- ▶ Average duration assumption: 2 years.
- ▶ Target saving: **0.01 bp** average yield-cost reduction.
- ▶ Annual P&L target:
 $45tn \times 2 \times 0.01 \times 10^{-4} = \text{RMB 90mm}$.

Yield saving	Approx. annual P&L
0.01 bp	RMB 0.09 bn
0.03 bp	RMB 0.27 bn
0.05 bp	RMB 0.45 bn
0.10 bp	RMB 0.90 bn

Message

Use **RMB 90mm** as the base-case annual P&L target. Upside comes from higher saving, larger covered flow, or longer duration exposure.

■ Roadmap and Leadership Ask

Two workstreams

- ▶ **Algo Type1:** 15–130 min direction signal plus fill-probability integration.
- ▶ **Internal flow:** internal crossing, routing, and liquidity selection engine.

Near-term milestones

- ▶ 2 weeks: data audit, baseline features, signal dashboard.
- ▶ 4–6 weeks: shadow trading and internal-flow simulator.
- ▶ 8–12 weeks: controlled production pilot.

Decision needed

- ▶ Confirm priority instruments and Algo instance.
- ▶ Approve access to LOB, order, fill, and internal flow data.
- ▶ Align on P&L attribution: slippage, market impact, and internal crossing savings.

**Deliver execution alpha first,
then scale it into a
department-level flow product.**